

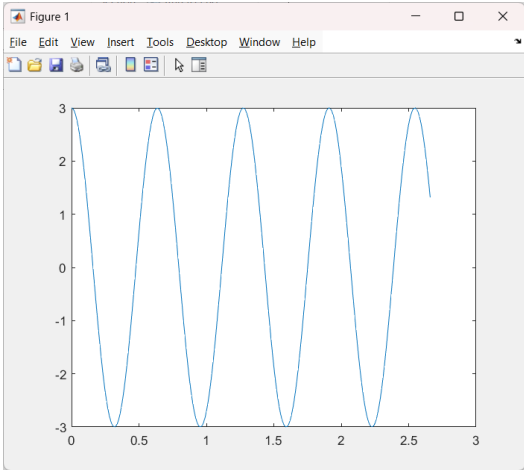
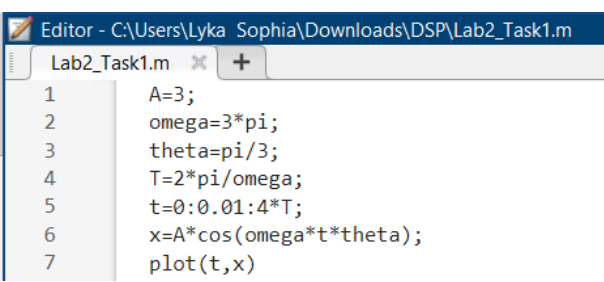


|          |                          |          |            |
|----------|--------------------------|----------|------------|
| Name:    | Lyka Sophia S. Jardeleza | Date:    | 08/22/2025 |
| Program: | BSCpE                    | Section: | A461       |

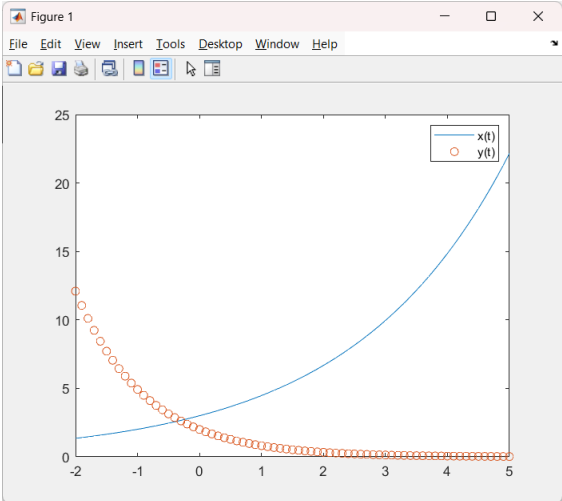
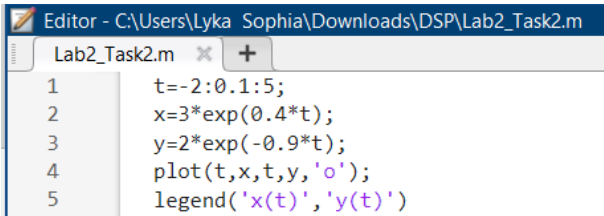
### A. Guided Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

#### 1 – Sinusoidal Signals

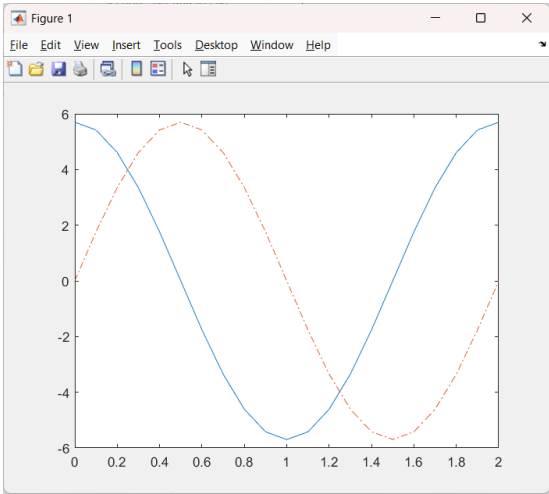
| Photo/s of MATLAB output                                                                            | Photos of MATLAB Codes                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 1</p>  | <p>a)</p>  <pre>1 A=3; 2 omega=3*pi; 3 theta=pi/3; 4 T=2*pi/omega; 5 t=0:0.01:4*T; 6 x=A*cos(omega*t*theta); 7 plot(t,x)</pre> |

#### 2 – Exponential Signals

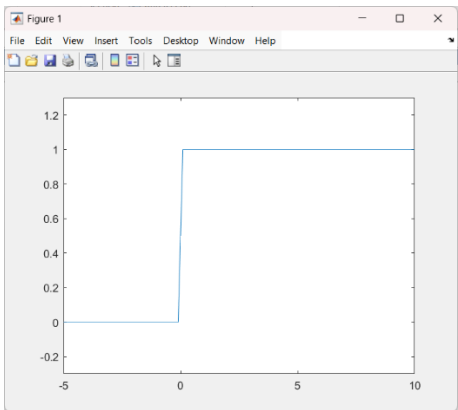
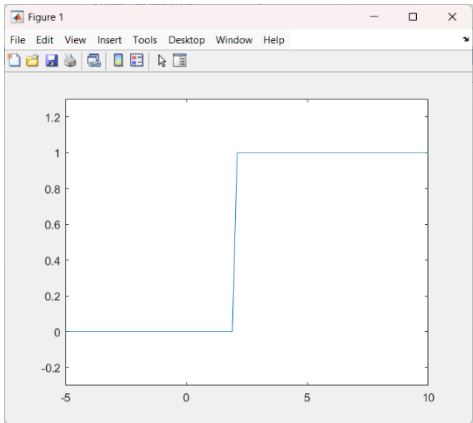
| Photo/s of MATLAB output                                                                             | Photos of MATLAB Codes                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 2</p>  | <p>a)</p>  <pre>1 t=-2:0.1:5; 2 x=3*exp(0.4*t); 3 y=2*exp(-0.9*t); 4 plot(t,x,t,y,'o'); 5 legend('x(t)','y(t)')</pre> |



### 3 – Complex Exponential Signals

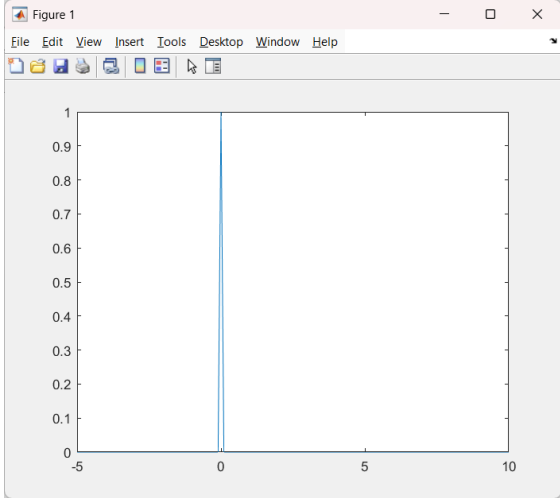
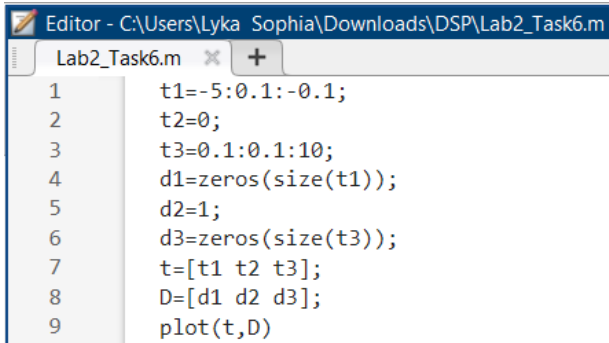
| Photo/s of MATLAB output                                                                           | Photos of MATLAB Codes                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 3</p>  | <p>a)</p> <pre>Editor - C:\Users\Lyka\Sophia\Downloads\DSP\Lab2_Task3.m Lab2_Task3.m 1 t=0:0.1:2; 2 y_re=real(2*exp(j*pi*t+pi/3)); 3 y_im=imag(2*exp(j*pi*t+pi/3)); 4 plot(t,y_re,t,y_im,'-.');</pre> |

### 4 – Unit Step Function

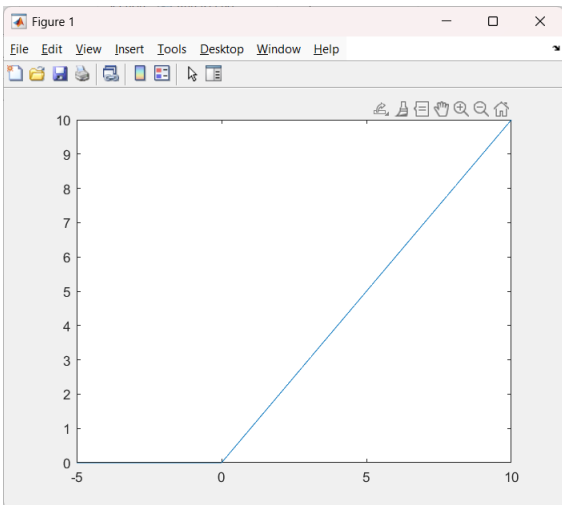
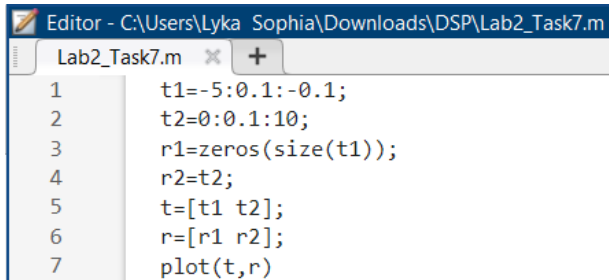
| Photo/s of MATLAB output                                                                                                                                                                                  | Photos of MATLAB Codes                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 4</p>  <p>b) Task 5</p>  | <p>a)</p> <pre>Editor - C:\Users\Lyka\Sophia\Downloads\DSP\Lab2_Task4.m Lab2_Task4.m 1 t=-5:0.1:10; 2 u=heaviside(t); 3 plot(t,u) 4 ylim([-0.3 1.3])</pre> <p>b)</p> <pre>Editor - C:\Users\Lyka\Sophia\Downloads\DSP\Lab2_Task5.m Lab2_Task5.m 1 t=-5:0.1:10; 2 u=heaviside(t-2); 3 plot(t,u) 4 ylim([-0.3 1.3])</pre> |



### 5 – Unit Impulse / Dirac Delta

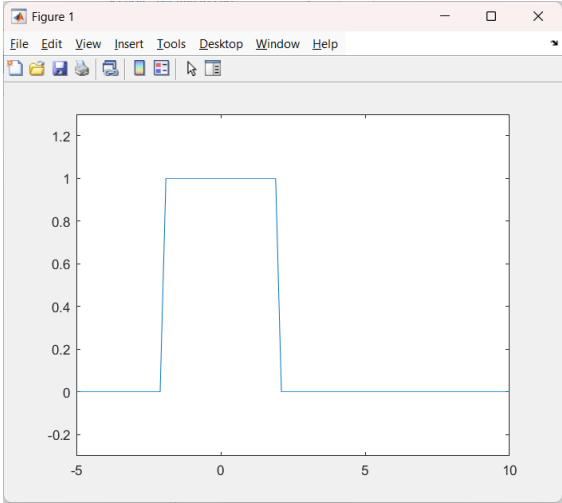
| Photo/s of MATLAB output                                                                           | Photos of MATLAB Codes                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 6</p>  | <p>a)</p>  <pre>1 t1=-5:0.1:-0.1; 2 t2=0; 3 t3=0.1:0.1:10; 4 d1=zeros(size(t1)); 5 d2=1; 6 d3=zeros(size(t3)); 7 t=[t1 t2 t3]; 8 D=[d1 d2 d3]; 9 plot(t,D)</pre> |

### 6 – Ramp Function

| Photo/s of MATLAB output                                                                             | Photos of MATLAB Codes                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 7</p>  | <p>a)</p>  <pre>1 t1=-5:0.1:-0.1; 2 t2=0:0.1:10; 3 r1=zeros(size(t1)); 4 r2=t2; 5 t=[t1 t2]; 6 r=[r1 r2]; 7 plot(t,r)</pre> |



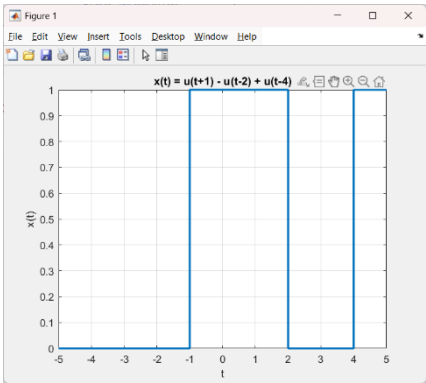
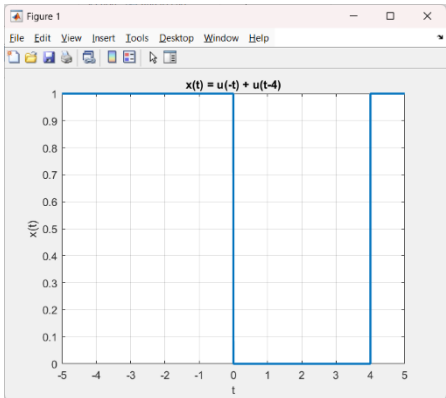
## 7 – Rectangular Pulse Function

| Photo/s of MATLAB output                                                                           | Photos of MATLAB Codes                                                                                                                                  |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a) Task 8</p>  | <p>a)</p> <pre> 1      t=-5:0.1:10; 2      u1=heaviside(t+2); 3      u2=heaviside(t-2); 4      p=u1-u2; 5      plot(t,p) 6      ylim([-0.3 1.3]) </pre> |

## B. Problem Exercises:

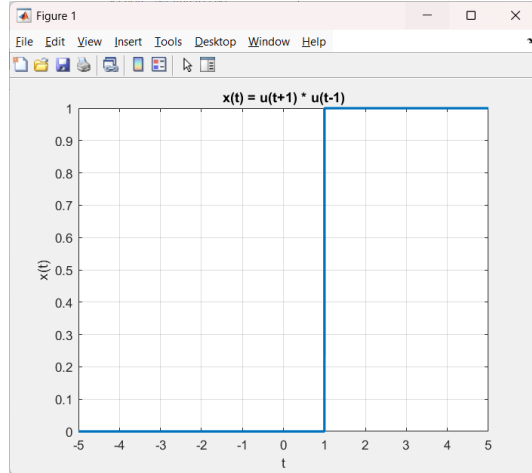
(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

### Problem 1:

| Photo/s of MATLAB output                                                                                                                                                                    | Photos of MATLAB Codes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>a)</p>  <p>b)</p>  | <p>a)</p> <pre> 1      % Exercise 1a 2      syms t 3      x = heaviside(t+1) - heaviside(t-2) + heaviside(t-4); 4 5      fplot(x, [-5, 5], 'LineWidth', 2) 6      grid on 7      xlabel('t') 8      ylabel('x(t)') 9      title('x(t) = u(t+1) - u(t-2) + u(t-4)') </pre> <p>b)</p> <pre> 1      % Exercise 1b 2      syms t 3      x = heaviside(-t) + heaviside(t-4); 4 5      fplot(x, [-5, 5], 'LineWidth', 2) 6      grid on 7      xlabel('t') 8      ylabel('x(t)') 9      title('x(t) = u(-t) + u(t-4)') </pre> |



c)

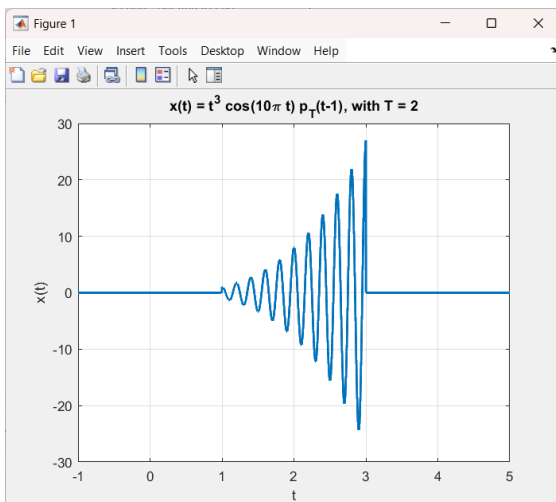


c)

```
Editor - C:\Users\Lyka Sophia\Downloads\DSP\Lab2_Exer1c.m
Lab2_Exer1c.m
1 % Exercise 1c
2 syms t
3 x = heaviside(t+1) * heaviside(t-1);
4
5 fplot(x, [-5, 5], 'LineWidth', 2)
6 grid on
7 xlabel('t')
8 ylabel('x(t)')
9 title('x(t) = u(t+1) * u(t-1)')
```

### Problem 2:

#### Photo/s of MATLAB output



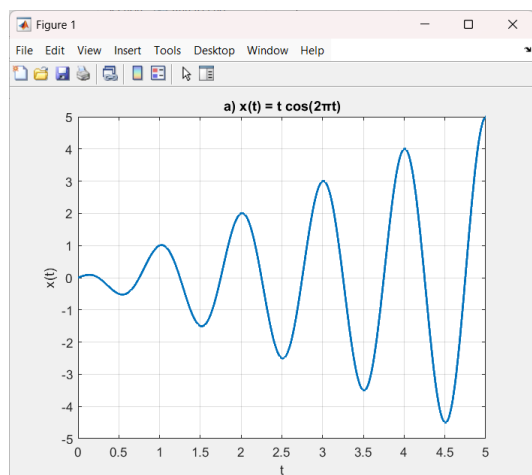
#### Photos of MATLAB Codes

```
Editor - C:\Users\Lyka Sophia\Downloads\DSP\Lab2_Exer2.m
Lab2_Exer2.m
1 % Exercise 2: Plot x(t) = t^3 cos(10*pi*t) p_T(t-1)
2
3 % Time axis
4 t = -1:0.001:5;
5
6 % Duration of pulse
7 T = 2;
8
9 % Rectangular pulse p_T(t-1) = u(t-1) - u(t-1-T)
10 pT = double((t >= 1) & (t <= 1+T));
11
12 % Define signal
13 x = (t.^3) .* cos(10*pi*t) .* pT;
14
15 % Plot
16 plot(t, x, 'LineWidth', 2)
17 grid on
18 xlabel('t')
19 ylabel('x(t)')
20 title(['x(t) = t^3 cos(10*pi t) p_T(t-1), with T = ', num2str(T)])
```

### Problem 3:

#### Photo/s of MATLAB output

a)

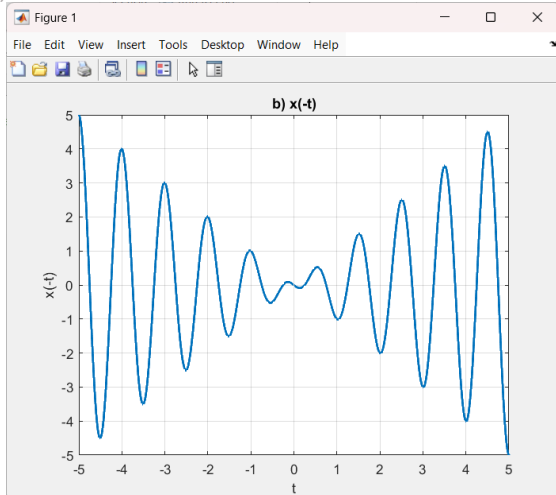


a)

```
Editor - C:\Users\Lyka Sophia\Downloads\DSP\Lab2_Exer3a.m
Lab2_Exer3a.m
1 % Exercise 3a
2 t = 0:0.001:5;
3 x = t .* cos(2*pi*t);
4
5 plot(t, x, 'LineWidth', 2)
6 grid on
7 xlabel('t')
8 ylabel('x(t)')
9 title('a) x(t) = t cos(2*pi*t)')
```



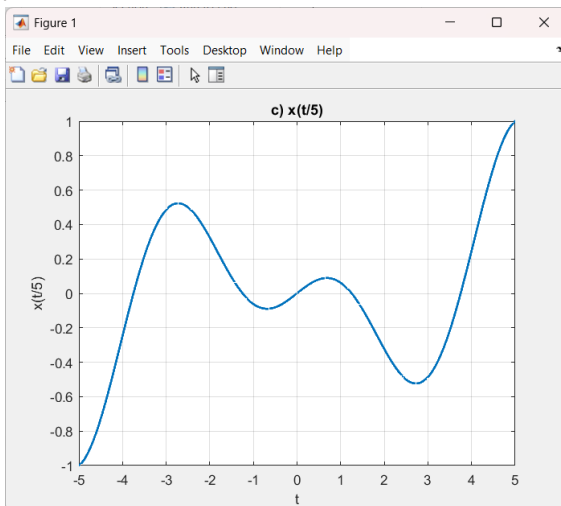
b)



b)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab2_Exer3b.m
Lab2_Exer3b.m
1 % Exercise 3b
2 t = -5:0.001:5; % symmetric domain to show time reversal
3 x = (-t) .* cos(2*pi*(-t));
4
5 plot(t, x, 'LineWidth', 2)
6 grid on
7 xlabel('t')
8 ylabel('x(-t)')
9 title('b) x(-t)')
```

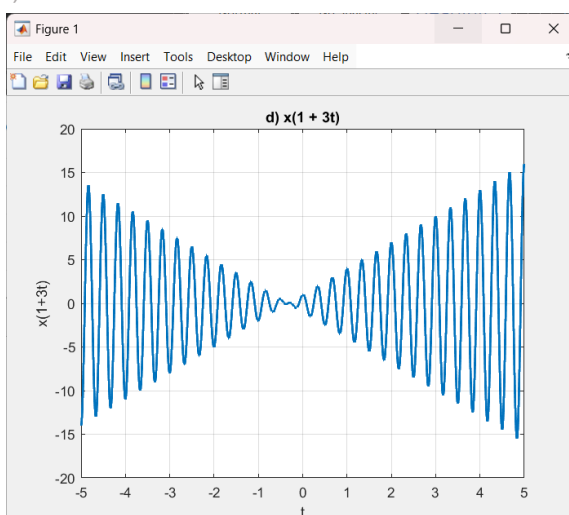
c)



c)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab2_Exer3c.m
Lab2_Exer3c.m
1 % Exercise 3c
2 t = -5:0.001:5;
3 x = (t/5) .* cos(2*pi*(t/5));
4
5 plot(t, x, 'LineWidth', 2)
6 grid on
7 xlabel('t')
8 ylabel('x(t/5)')
9 title('c) x(t/5)')
```

d)

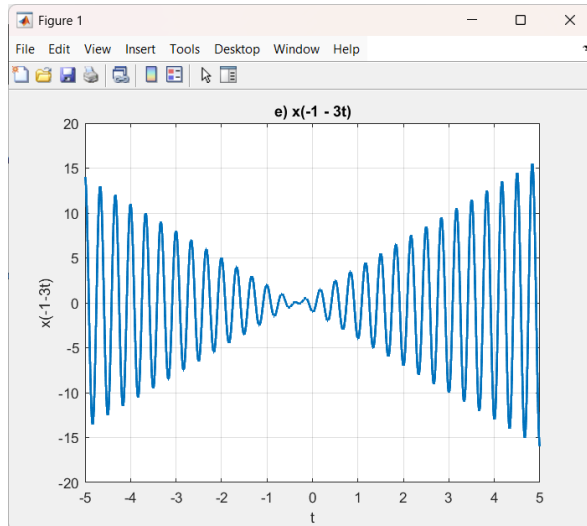


d)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab2_Exer3d.m
Lab2_Exer3d.m
1 % Exercise 3d
2 t = -5:0.001:5;
3 x = (1+3*t) .* cos(2*pi*(1+3*t));
4
5 plot(t, x, 'LineWidth', 2)
6 grid on
7 xlabel('t')
8 ylabel('x(1+3t)')
9 title('d) x(1+3t)')
```



e)



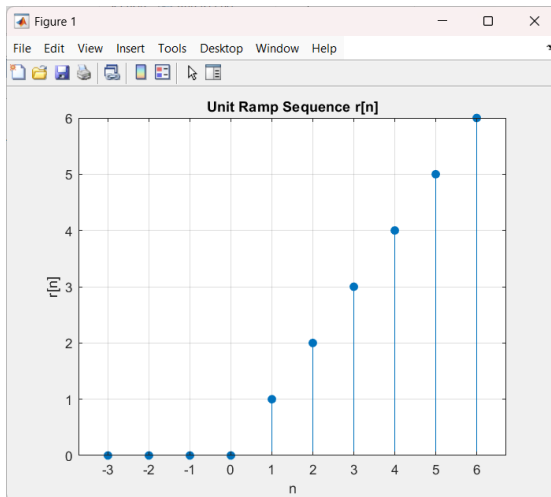
e)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab2_Exer3e.m
Lab2_Exer3e.m
1 % Exercise 3e
2 t = -5:0.001:5;
3 x = (-1-3*t) .* cos(2*pi*(-1-3*t));
4
5 plot(t, x, 'LineWidth', 2)
6 grid on
7 xlabel('t')
8 ylabel('x(-1-3t)')
9 title('e) x(-1 - 3t)')
```

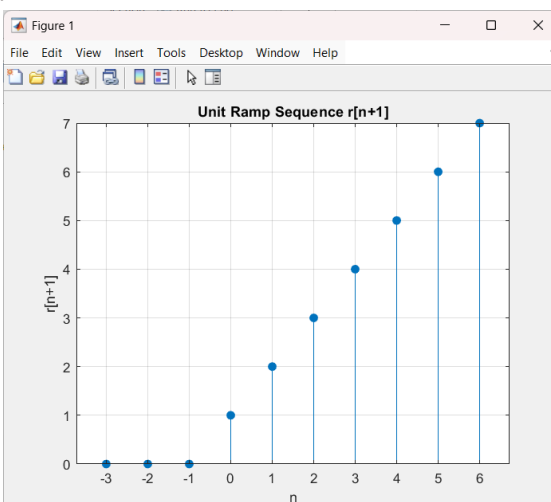
#### Problem 4:

##### Photo/s of MATLAB output

a)



b)



##### Photos of MATLAB Codes

a)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab2_Exer4a.m
Lab2_Exer4a.m
1 % (a) r[n] in the interval -3 <= n <= 6
2 n = -3:6; % time index
3 r = n .* (n >= 0); % ramp definition: r[n] = n if n>=0 else 0
4
5 stem(n, r, 'filled');
6 xlabel('n');
7 ylabel('r[n]');
8 title('Unit Ramp Sequence r[n]');
9 grid on;
```

b)

```
Editor - C:\Users\Lyka_Sophia\Downloads\DSP\Lab2_Exer4b.m
Lab2_Exer4b.m
1 % (b) r[n+1] in the interval -3 <= n <= 6
2 n = -3:6; % time index
3 r_shifted = (n+1) .* (n >= -1); % ramp shifted: r[n+1] = n+1 if n>=-1 else 0
4
5 stem(n, r_shifted, 'filled');
6 xlabel('n');
7 ylabel('r[n+1]');
8 title('Unit Ramp Sequence r[n+1]');
9 grid on;
```



## **SIGNALS**

### **C. Findings ,Observations, and Comments:**

#### *Guiding Questions :*

1. *What did you learn in the guided experiments? Did you learn something new?*
2. *What are your findings, observations, and comments in the guided exercises?*
3. *Did you solve the problem exercises?*
4. *Are there any difficulties during coding or set-up that you encountered?*
5. *What are your findings, observations, and comments in the problem exercise?*

In the guided experiments, I learned how to represent and plot basic digital signals in MATLAB like sinusoidal, exponential, step, impulse, ramp, and rectangular pulses. MATLAB made it easier to visualize these compared to just equations, and I saw how each signal behaves, such as the step starting at zero, the ramp increasing linearly, and exponentials either growing or decaying.

In the guided exercises, I noticed that small changes in equations greatly affect the graphs. For sinusoidal signals, amplitude, frequency, and phase shift change the wave's appearance. For exponentials, positive exponents grow while negatives decay. Step and rectangular pulses acted like switches, turning on and off at specific points, which helped me connect math with visual results.

I was able to solve the problem using MATLAB functions like heaviside, exp, and array operations. At first, applying shifts and scaling was confusing, but after practice, I understood how to transform functions correctly.

I also faced difficulties with MATLAB syntax, like missing operators or parentheses, and understanding time shifts and scaling. My plots sometimes didn't match expectations, so I had to recheck equations. Representing impulses and rectangular pulses was tricky at first, but I learned how to do it properly in MATLAB.

In the problem exercises, I saw how transformations like shifting, scaling, and time reversal change signals. For example, Exercise 3 showed different versions of the same signal under transformations, while Exercise 4 showed that the ramp and its shifted version are similar but shifted in time. Overall, the tasks improved my understanding of signals and boosted my confidence in using MATLAB for DSP.